Roadmap of the day

Start and stop. Each exercise starts when its slide is on screen and stops when its time is up: the instructor calls time and moves on. Unfinished work stays unfinished; the key points come in the debrief.

"Pack" = page of this pack. Files: github.com/ielbadisy/critical-appraisal-workshop

#	Step	Min	Slides	Pack	Also on your desk
1	Welcome and goals	5	1-3		
2	Pre-quiz	10	4	p. 2-3	
3	Talk: open science and pre-specification	25	5-11		
4	**Exercise 1** (pairs): registry to paper; debrief with the annotated papers	60	12-13	p. 4-5	Both papers; materials/pdfs/; registries online
	Break	15	14		
5	Talk: statistical methodology	20	15-24		
6	**Exercise 2** (groups): four stations; debrief with the annotated papers	4x20	25-26	p. 6-7; toolkit, p. 23-25	Both papers, annotated
	Lunch	60	27		
7	Briefing: apply the method, then ask an LLM	5	28-30		
8	**Exercise 3** (alone, no AI): review manuscript A	50	31	p. 8; manuscript A, p. 9-12; checklist, p. 19-22	
9	**Exercise 4** (pairs): merge lists, write the report	15	32-33	p. 13; report template, p. 14-15	
10	**Exercise 5** (alone): ask an LLM with prompt G or S	10	34-35	p. 16	llm-exercise/manuscript-A.md
11	**Exercise 6** (alone, then pairs): audit every LLM point	25	36	p. 17	
	Break	15	37		
12	Talk: LLMs in peer review	20	38-45		
13	**Exercise 7** (plenary): reveal and compare	55	46-48	p. 18	Annotated manuscript A
14	Wrap-up and post-quiz	15	49-51	p. 26-27	

Keep after the day: review report template (p. 14-15), 7-layer checklist (p. 19-22), toolkit (p. 23-25).

Pre-quiz

The same 15 questions are taken at the start (pre) and at the end (post) of the day. One answer per question. About 10 minutes.

Participant code: _____ Pre-quiz

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1. A randomised trial finds RR 0.95 (95% CI 0.80 to 1.13) for its primary outcome. The best conclusion is:

- A. The treatment has no effect.
- B. The trial did not show a difference; effects from a 20% reduction to a 13% increase remain compatible with the data.
- C. The treatment is equivalent to control.
- D. The trial was underpowered, so the result should be ignored.

2. A trial's primary outcome in the registry is "mortality at 90 days"; the paper reports "mortality at 28 days" as primary. This is:

- A. Acceptable if the 28-day result is significant.
- B. Outcome switching, which should be disclosed and justified.
- C. A minor reporting issue.
- D. Only a problem if the journal requires registration.

3. The main purpose of a statistical analysis plan (SAP) finalised before unblinding is to:

- A. Calculate the sample size.
- B. Satisfy the ethics committee.
- C. Limit analytical flexibility after the data are seen.
- D. Describe the randomisation procedure.

4. In a trial, 12 randomised patients are excluded from the analysis because they did not receive the intervention. The analysis is described as intention-to-treat. This is:

- A. Correct, because they did not receive the treatment.
- B. Incorrect: intention-to-treat analyses patients in the group they were randomised to.
- C. Correct if the exclusions are balanced between groups.
- D. Correct if the exclusions are reported in the flow diagram.

5. A paper reports OR 1.40 (95% CI 1.05 to 1.87) and p = 0.21. You should:

- A. Trust the p-value.
- B. Trust the CI.
- C. Flag an internal inconsistency: a 95% CI excluding 1 implies $p < 0.05$.
- D. Ignore it; both are correct with different tests.

6. In a subgroup of n = 17 patients, the mean number of previous pregnancies (an integer variable) is reported as 2.30. This is:

- A. Plausible.
- B. Impossible, because 2.30 x 17 = 39.1 is not close to a whole number.
- C. Only checkable with the raw data.
- D. Possible if some values are missing.

7. Which design can best support a causal claim about a drug's effect?

- A. A large registry cohort with a very small p-value.
- B. A well-conducted randomised controlled trial.
- C. A case series of consecutive patients.
- D. A cross-sectional survey with adjustment for 50 covariates.

8. The reporting guideline for a study developing a machine-learning prediction model is:

- A. CONSORT-AI
- B. STARD
- C. TRIPOD+AI
- D. STROBE

9. The risk-of-bias tool for a non-randomised study comparing two treatments is:

- A. RoB 2
- B. ROBINS-I
- C. QUADAS-2
- D. GRADE

10. The primary outcome is non-significant, and the abstract conclusion highlights a significant subgroup. This is an example of:

- A. Good transparency.
- B. Spin.
- C. Multiplicity correction.
- D. A pre-specified analysis, so not a problem.

11. Which is a typical signal of a paper-mill product?

- A. Data shared on a public repository.
- B. Tortured phrases such as “counterfeit consciousness” for “artificial intelligence”.
- C. A long limitations section.
- D. Registration before enrolment.

12. You are reviewing a confidential manuscript for a journal. Regarding AI tools:

- A. You may upload the full manuscript to any LLM to save time.
- B. You must not upload the confidential manuscript; follow the journal’s policy and disclose any permitted use.
- C. You may upload it if you delete the chat afterwards.
- D. AI use is never relevant to peer review.

13. An LLM tells you that a trial’s registry entry lists a different primary outcome. The best next step is:

- A. Report it as a major issue.
- B. Ignore it.
- C. Open the registry yourself and verify, including the history of changes.
- D. Ask the LLM again to confirm.

14. Unblinded outcome assessment is most likely to bias:

- A. All-cause mortality.
- B. A subjective outcome such as a functional score rated by the treating team.
- C. Neither: randomisation protects against it.
- D. Only the baseline characteristics.

15. In a review report, a major issue should ideally contain:

- A. Your general impression of the authors’ competence.
- B. The problem, why it matters, and a concrete request the authors can act on.
- C. Only the recommendation to reject.
- D. A list of typos.

Exercise 1. Open science: what was promised, what was published

When: block 1, 60 minutes. **Format:** pairs. Laptops open.

You trace two real papers from their registration to their publication: one well-run trial and one retracted study. Then you score both on the same open-science scorecard.

Sources

	TTM2 (Dankiewicz et al., *N Engl J Med* 2021)	Gautret et al., *Int J Antimicrob Agents* 2020 (retracted 2024)
Registry	NCT02908308 ("History of changes" tab)	EU Clinical Trials Register 2020-000890-25
Protocol	*Am Heart J* 2019 (materials/pdfs/article-ttm2-protocol-2019.pdf)	Not published; the registry entry is the only public plan
SAP	*Trials* 2020 (materials/pdfs/article-ttm2-sap-2020.pdf)	None
Paper	NEJM 2021 and supplementary appendix	Printed copy (paywalled)

Part A. Trace TTM2 (25 minutes)

Item	Registry (first version)	Protocol 2019	SAP 2020	Paper 2021	Consistent?
Primary outcome and time point					
Main secondary outcome					
Planned sample size					
Assumed control-group risk and effect size					
Primary analysis population					
Handling of missing data					
Date					

1. Was the trial registered before the first patient was enrolled? Give both dates.
2. Was the SAP published before the data were unblinded? How can you tell?
3. The sample size increased during the trial. When, why, and was it decided before any results were known?

Part B. Trace Gautret (20 minutes)

Item	Registry entry	Paper	Consistent?
Design (randomised? controlled? blinded?)			
Planned sample size			
Primary end point and time point			
Control group			
Analysis population			

4. The paper says the results are "those of 36 patients" out of 42 enrolled, and that the authors shared them early "because our first results are so significant and evident". What is this analysis, in the language of a protocol? Was a rule for it pre-specified?
5. Six treated patients are not in the analysis. Which document should have said what happens to them?
6. The paper was accepted about one day after submission, in a journal where an author was an editor. Which layer is this, and what does it change for you as a reader?

Part C. Outcome switching at scale (5 minutes, read and discuss)

COMPare (Goldacre et al., *Trials* 2019; materials/pdfs/method-goldacre-compare-2019.pdf) checked every trial published over six weeks in NEJM, Lancet, JAMA, BMJ and Annals: 67 trials.

7. In how many trials did the published outcomes differ from the pre-specified ones? What share of correction letters did the journals publish?
8. What is the simplest check a reviewer can do in five minutes to catch outcome switching?

Part D. Open-science scorecard (10 minutes)

Score each paper: **Yes / Partly / No / Cannot tell**.

Item	TTM2	Gautret
Registered before first enrolment		
Protocol public before results		
SAP public before unblinding		
Primary outcome unchanged from registry to paper		
All registered outcomes reported		
Data-sharing statement		
Individual patient data actually obtainable		
Analysis code available		
Independent peer review (no editorial conflict)		

9. The Elgazzar ivermectin preprint was withdrawn after independent readers found duplicated patients and deaths dated before the study started. They could do this only because the **raw data** became available. Which row of the scorecard would have protected readers, and why does peer review almost never get there?

Debrief. At the end you receive the annotated TTM2 and Gautret papers: every point of Parts A, B and D is marked on the page. Check your answers against them.

Exercise 2. Statistical methodology on real papers

When: block 2, 80 minutes. **Format:** groups of 3-4, rotating through four stations (20 minutes each). A calculator or phone is enough. Formulas are in the toolkit.

All four stations use the two morning papers: **TTM2** (randomised) and **Gautret 2020** (non-randomised, retracted).

Station 1. The estimand: what exactly is being estimated?

1. Write the TTM2 primary estimand in five parts:

Attribute	TTM2
Population	
Treatment strategies compared	
Outcome variable and time point	
Intercurrent events (patients not cooled as planned; withdrawal of life support) and how they are handled	
Population-level summary measure	

2. Try the same for Gautret. Which attributes can you fill, and which are missing or change during the paper?
3. Gautret compares treated patients in one centre with untreated patients from other centres and patients who refused. What question does this comparison actually answer?
4. The TTM2 SAP assumes 55% control mortality and an absolute reduction of 7.5 points. Recompute the relative reduction. The paper says "15%". Which is right, and does it matter?

Station 2. Confidence intervals and p-values

1. TTM2 reports death 465/925 vs 446/925, RR 1.04 (95% CI 0.94 to 1.14), P = 0.37. Recompute the two percentages.
2. From the CI, estimate the p-value: SE = (ln upper - ln lower) / 3.92, then z = ln(RR) / SE. Is it compatible with 0.37?
3. Write one sentence that describes this result **correctly**. Then write the most common **wrong** sentence a reader would write.
4. What range of effects is compatible with the TTM2 data? Would a 6% relative reduction in death be ruled out?
5. Gautret reports p-values on 36 patients, one per day from day 3 to day 6, for several comparisons. How many tests do you count? What does that do to a p = 0.001?

Station 3. Do the numbers add up? (forensics)

A. TTM2 flow. The abstract says 1900 enrolled and 1850 evaluated. The ITT population is 1861 (930 and 931). Account for every patient from 1900 to 1861 to 1850. Why do the arrhythmia and functional-outcome rows use other denominators, and is that a problem?

B. Gautret flow. 42 enrolled, 36 analysed, 26 treated, 16 controls. Rebuild the flow diagram that the paper does not give. Who is missing, from which arm, and why?

C. Quick checks (fictional numbers).

1. Subgroup n = 13, mean number of admissions 2.41. Possible? (GRIM)
2. OR 1.32 (95% CI 1.01 to 1.73), p = 0.11. Consistent?
3. Table 1: 87 of 120 women (75.5%). Check.
4. A trial of 200 patients reports 22 baseline p-values, all between 0.62 and 0.98. Why might this be suspicious?

Station 4. Bias and spin

A. RoB 2 on TTM2. Concealed web-based randomisation; clinicians not blinded; blinded assessors for functional outcome; primary outcome death; published SAP.

RoB 2 domain	Low / Some concerns / High	Why
1. Randomisation process		
2. Deviations from intended interventions		
3. Missing outcome data		
4. Measurement of the outcome		
5. Selection of the reported result		

B. ROBINS-I on Gautret.

ROBINS-I domain	Low / Moderate / Serious / Critical	Why
1. Confounding		
2. Selection of participants		
3. Classification of interventions		
4. Deviations from intended interventions		
5. Missing data		
6. Measurement of outcomes		
7. Selection of the reported result		
Overall		

C. Spin. Read the Gautret abstract conclusion. Underline every claim the design cannot support. Use the spin types from Boutron et al. (*JAMA* 2010): focus on a secondary or subgroup result, "trend" wording, causal language from a non-randomised design, recommendation beyond the data. Rewrite the conclusion honestly in at most 40 words.

_____

Group summary (5 minutes, at the end)

For each station, write the single most important methodological point in one sentence, as you would put it in a review report.

Debrief. Keep the annotated TTM2 and Gautret papers from Exercise 1 at hand: the station answers (estimand, recomputed p-values, forensics, bias and spin) are marked on the page.

Exercise 3. Review manuscript A alone

Format: individual, 50 minutes, no AI. **Use:** manuscript A and the 7-layer checklist (this pack).

This afternoon you apply the morning's method to one new paper. Manuscript A is **fictional**: a short trial report with its registry entry and SAP excerpt, written for training. Because it is fictional and not confidential, you may paste it into an LLM in Exercise 5. **You must never do this with a real manuscript you review for a journal.**

Work through the seven layers in order. For each problem, give its location, what is wrong (show your arithmetic) and why it matters.

#	Layer	Location	What is wrong	Why it matters
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

FICTIONAL TEACHING MANUSCRIPT A

> This manuscript, its registry entry and its analysis plan are entirely fictional. They were written for a peer-review training exercise and contain deliberate errors. Do not cite.

Your task: you are a peer reviewer. Read the registry entry, the SAP excerpt and the manuscript. On the Exercise 3 sheet of the participant pack, list every problem you find, with its location and why it matters.

Document 1. Registry entry (excerpt)

Field	Content
Registration number	REG-2021-0147
Date registered	2 March 2021
Date of first enrolment (planned)	1 April 2021
Title	Early mobilisation after hip fracture surgery (EMOB-HIP): a multicentre randomised trial
Design	Parallel-group, two-arm, 1:1, randomised, assessor-blinded
Population	Adults aged 65 or older after surgery for proximal femoral fracture
Intervention	Physiotherapist-led mobilisation (standing and walking) started within 24 hours of surgery
Comparator	Usual care: mobilisation started at the discretion of the ward team, usually day 2 or 3
Primary outcome	Independent walking, defined as Functional Ambulation Category (FAC) of 4 or more, at day 30 after surgery. Assessed by a physiotherapist blinded to allocation.
Secondary outcomes	FAC 4 or more at hospital discharge; delirium (4AT) during hospital stay; length of stay; falls during stay; 30-day mortality
Sample size	320 (160 per arm)

Document 2. Statistical analysis plan (excerpt, version 1.0, signed 15 March 2021)

Sample size. Assuming 40% of usual-care patients walk independently at day 30, 153 patients per arm give 80% power to detect an increase to 56% (two-sided alpha 0.05). Allowing for 5% loss to follow-up, we will randomise 160 per arm (320 in total).

Analysis population. The primary analysis will follow the intention-to-treat principle: all randomised patients will be analysed in the group to which they were allocated, whether or not they received the allocated intervention. Patients with a missing primary outcome will be excluded from the primary analysis; a sensitivity analysis will use multiple imputation.

Primary analysis. The primary outcome will be compared using a risk ratio with 95% confidence interval and a chi-squared test.

Secondary outcomes. Analysed in the same way, without adjustment for multiplicity; they will be interpreted as supportive.

Document 3. Manuscript

Early mobilisation within 24 hours after hip fracture surgery: a multicentre randomised controlled trial

Abstract

Background. Delayed mobilisation after hip fracture surgery is associated with poor recovery. We tested whether mobilisation within 24 hours improves independent walking.

Methods. In this multicentre randomised trial in four hospitals, patients aged 65 or older were randomly assigned to physiotherapist-led mobilisation within 24 hours of surgery or usual care. The primary outcome was independent walking (FAC 4 or more) at hospital discharge.

Results. 324 patients were randomised. Independent walking at discharge was achieved by 89 of 152 patients (58.6%) in the early mobilisation group and 64 of 154 (41.6%) in the usual-care group (RR 1.41, 95% CI 1.12 to 1.77; p = 0.003). Delirium was less frequent with early mobilisation (20.4% vs 29.2%; p = 0.02).

Conclusion. Early mobilisation restores independent walking after hip fracture surgery and should become standard care.

Introduction

Hip fracture is a leading cause of disability in older adults. Fewer than half of patients regain their pre-fracture walking ability. Observational studies suggest that mobilisation on the day after surgery is associated with better function and lower mortality, but these studies are prone to confounding because healthier patients are mobilised earlier. Guidelines recommend early mobilisation, but randomised evidence is limited. We conducted a pragmatic randomised trial to test whether physiotherapist-led mobilisation within 24 hours of surgery improves independent walking.

Methods

Design and setting. EMOB-HIP was a parallel-group randomised trial in four hospitals, registered before enrolment (REG-2021-0147). The ethics committee approved the protocol, and all patients or their legal representatives gave written informed consent.

Participants. Patients aged 65 or older undergoing surgery for a proximal femoral fracture were eligible. Patients who were non-ambulatory before the fracture, or had a pathological fracture, were excluded.

Randomisation. After surgery, patients were randomised 1:1 using a web-based system with permuted blocks of variable size, stratified by site. Allocation was concealed until assignment.

Interventions. In the early group, a physiotherapist helped the patient to stand and walk with an aid within 24 hours of the end of surgery, then twice daily. In the usual-care group, mobilisation started at the discretion of the ward team.

Outcomes. The primary outcome was independent walking at hospital discharge, defined as an FAC of 4 or more. FAC was assessed by the ward physiotherapist responsible for the patient's rehabilitation. Secondary outcomes were FAC 4 or more at day 30, delirium (4AT score of 4 or more on any day), length of stay, falls during the stay and 30-day mortality.

Sample size. We planned to randomise 320 patients (see the published SAP).

Statistical analysis. All analyses followed the intention-to-treat principle. Binary outcomes were compared with risk ratios, 95% confidence intervals and chi-squared tests. Missing outcomes were addressed with multiple imputation in a sensitivity analysis.

Results

Participants. Between April 2021 and September 2023, 160 patients were randomised to early mobilisation and 160 to usual care. In the early group, 8 patients were excluded after randomisation because mobilisation could not be started within 24 hours (haemodynamic instability in 5, pain in 3). In the usual-care group, 6 patients were lost to follow-up. The intention-to-treat analysis therefore included 152 and 154 patients. Baseline characteristics were well balanced (Table 1).

Table 1. Baseline characteristics

Characteristic	Early mobilisation (n = 160)	Usual care (n = 160)
Age, years, mean (SD)	82.4 (7.1)	82.9 (6.8)
Female, n (%)	118 (71.5)	115 (71.9)
Independent walking before fracture, n (%)	141 (88.1)	139 (86.9)
Dementia, n (%)	19 (11.9)	21 (13.1)
Charlson comorbidity index, mean (SD)	2.1 (1.6)	2.2 (1.7)
Intracapsular fracture, n (%)	92 (57.5)	95 (59.4)
ASA grade 3 or 4, n (%)	101 (63.1)	97 (60.6)

In the early group, patients with dementia (n = 19) had a mean Charlson comorbidity index of 3.48.

Primary outcome. At discharge, 89 of 152 patients (58.6%) in the early group walked independently, compared with 64 of 154 (41.6%) in the usual-care group (RR 1.41, 95% CI 1.12 to 1.77; p = 0.003).

Table 2. Outcomes

Outcome	Early mobilisation (n = 152)	Usual care (n = 154)	RR (95% CI)	p
Primary: FAC 4 or more at discharge	89 (58.6)	64 (41.6)	1.41 (1.12 to 1.77)	0.003
FAC 4 or more at day 30	82 (53.9)	72 (46.8)	1.15 (0.92 to 1.44)	0.21
Delirium during stay	31 (20.4)	45 (29.2)	0.70 (0.47 to 1.04)	0.02
Fall during stay	7 (4.6)	5 (3.2)	1.42 (0.46 to 4.37)	0.54
Death by day 30	9 (5.9)	11 (7.1)	0.83 (0.35 to 1.94)	0.67
Length of stay, days, median (IQR)	9 (7 to 13)	11 (8 to 15)		0.01

The multiple-imputation sensitivity analysis gave similar results.

Discussion

In this randomised trial, mobilisation within 24 hours of hip fracture surgery increased independent walking at discharge by 41% and reduced delirium. These findings are consistent with observational studies and support current guideline recommendations. The effect on delirium is biologically plausible, as early mobilisation reduces immobility, sleep disturbance and the use of urinary catheters.

Strengths of our trial include randomisation with concealed allocation, a pre-registered protocol and a published analysis plan. Limitations include the impossibility of blinding patients and physiotherapists, and the conduct of the trial in a single country.

Conclusion. Early mobilisation restores independent walking after hip fracture surgery and should become standard care for all patients.

Exercise 4. Merge in pairs and write the report

Format: pairs, 15 minutes. **Use:** your Exercise 3 list and the review report template (this pack).

Merge your two lists. What did one of you see that the other missed, and in which layer? Write the review report with the template: summary, major issues, minor issues, recommendation.

Peer-Review Report Template

Manuscript: _____ **Reviewer:** _____ **Date:** _____

1. Summary (3-5 sentences)

What the study asked, how, what it found, and what the authors conclude. Write it neutrally: the authors should recognise their own paper.

2. Overall assessment (2-3 sentences)

The single most important strength and the single most important problem, and whether the conclusions are supported.

3. Major issues

Problems that could change the conclusions. Number them. For each issue:

- **What:** the problem, with location (page, table, line).
- **Why it matters:** how it could affect the result or its interpretation, and in which direction.
- **What to do:** a concrete request (an analysis, a clarification, a correction, or softer wording).

Organise them by the 7 layers when useful: design, pre-specification, internal consistency, external consistency, validity, interpretation.

1.
2.
3.

4. Minor issues

Reporting gaps, unclear wording, typos in numbers that do not change conclusions, missing checklist items (CONSORT, STROBE…).

1.
2.

5. Recommendation (for the editor only)

☐ Accept
☐ Minor revision
☐ Major revision
☐ Reject

Confidence in the main conclusion (verdict rule of the 7-layer checklist): high / moderate / low / very low

6. Confidential comments to the editor (optional)

Concerns about integrity (possible data fabrication, duplicate publication, undisclosed conflicts) belong here, stated factually.

7. AI use disclosure

☐ No AI tool was used.
☐ An AI tool was used for: _____. No confidential content was uploaded.

Tone checklist

- ☐ Every criticism is specific and located.
- ☐ Every major issue comes with a request the authors can act on.
- ☐ No comment on the authors as people; only on the work.
- ☐ Strengths are acknowledged.

Exercise 5. Ask an LLM

Format: individual, laptop, 10 minutes. **Use:** the file llm-exercise/manuscript-A.md.

Use any LLM you have access to. The instructor assigns your prompt: **odd-numbered seats use prompt G, even-numbered seats use prompt S.** Paste the prompt, then the manuscript file **from "Document 1" to the end** (registry entry, SAP excerpt and manuscript; leave out the title box above it). Note the model name and version.

Prompt G (generic):

> You are a peer reviewer for a general medical journal. Review the paper below.

Prompt S (structured):

> You are a peer reviewer for a general medical journal. Review the paper below by working through these components in order. Each component feeds the next.
>
> 1. Article type and guideline. Name the study design and the reporting guideline that applies (CONSORT for randomised trials, STROBE for observational studies, PRISMA for systematic reviews, STARD for diagnostic accuracy, TRIPOD+AI for prediction models). Use its items as your checklist.
> 2. External consistency. Compare the paper with every source provided (registry, protocol, analysis plan): primary outcome and its time point, design, blinding, sample size, analysis population. If a source is not provided, say which checks you could not do.
> 3. Internal consistency. Recompute every number you can: participant flow, percentages from n/N, confidence intervals against p-values, means of integer data, and the abstract against the text and tables.
> 4. Design and risk of bias. Randomisation and concealment, blinding of outcome assessment, deviations from the intended interventions, missing data, and whether the analysis population matches its stated definition.
> 5. Claims and language. Is each claim supported by the primary result? Check causal language, spin, relative versus absolute effects, harms and generalisation.
> 6. Integration. Link the problems: say how each one changes the credibility of the main result and of the conclusion.
>
> For each problem give its location, what is wrong (show your arithmetic), and why it matters. Rank problems by importance. Do not invent information that is not in the documents, and mark anything you could not verify.

Prompt used: G / S. Model and version: _____

Exercise 6. Audit every LLM point

Format: individual, then pairs, 25 minutes. **Use:** the LLM output from Exercise 5 and your Exercise 3 list.

For each point the LLM raised, **check it yourself in the documents** before you classify it.

LLM point (short)	Already on my list?	Verified true?	Classification
	Y / N	Y / N	Added / Duplicate / Wrong / Vague
	Y / N	Y / N	Added / Duplicate / Wrong / Vague
	Y / N	Y / N	Added / Duplicate / Wrong / Vague
	Y / N	Y / N	Added / Duplicate / Wrong / Vague
	Y / N	Y / N	Added / Duplicate / Wrong / Vague
	Y / N	Y / N	Added / Duplicate / Wrong / Vague
	Y / N	Y / N	Added / Duplicate / Wrong / Vague
	Y / N	Y / N	Added / Duplicate / Wrong / Vague

- **Added**: true, and you had missed it.
- **Duplicate**: true, and you had found it.
- **Wrong**: not supported by the documents (a misread number, an invented registry detail, a wrong calculation, an invented reference).
- **Vague**: generic advice that applies to any trial (“consider confounding”, “sample size may be small”).

Exercise 7. Reveal and compare

Format: plenary, 55 minutes. **Use:** your Exercises 3 to 6.

The instructor shows the planted errors and hands out the annotated manuscript A. Score yourself first, then the LLM.

	Me alone	Pair	LLM (prompt G or S)	Me + LLM, after audit
Planted errors found				
Wrong claims				

The instructor then shows the same comparison from repeated LLM runs: generic versus structured prompt, with and without the registry and SAP.

Discussion:

1. Which layers did you cover well, and which did you skip?
2. Errors that need **arithmetic** (a percentage, a p-value from a CI, an impossible mean): who found them more often, you or the LLM?
3. Errors that need a **second document** (registry, SAP): what can any reviewer, human or LLM, find without it?
4. Prompt G versus prompt S: what did the structure change? Where did the structure come from?
5. Did the LLM state anything with confidence that was false? How long did it take you to check?
6. Your journal's policy: may a reviewer use an LLM? Under which conditions, and what must be disclosed?

One rule

Write the one rule about AI in peer review that you will keep:

_____

To go further

The reporting guidelines are where each component of the review is defined in full. Start with the one for the designs you read most: see Layer 1 of the toolkit.

7-Layer Appraisal Checklist

Paper: _____ Reviewer: _____ Date: _____

How to use it. Work through the layers in order: each one feeds the next. Answer every item **Y** (yes), **N** (no) or **?** (cannot tell from the documents), and write where you saw it (page, table, line, registry field). Then rate the layer, and apply the verdict rule at the end. For a randomised trial, use the CONSORT annex alongside layers 1 to 4.

Layer rating: **OK / Concern / Serious concern** (could change the main result or its interpretation).

1. Article type

Item	Y/N/?	Where
1.1 The design is identified (RCT, cohort, case-control, cross-sectional, diagnostic, prediction/ML, SR/MA)		
1.2 The claim type is clear: descriptive, associational, causal or predictive		
1.3 The design can support that claim type		
1.4 The matching reporting guideline is named (CONSORT, STROBE, STARD, TRIPOD+AI, PRISMA)		

Rating: OK / Concern / Serious concern

2. Question and design

Item	Y/N/?	Where
2.1 The question fits PICO/PECO and the estimand is stated or deducible		
2.2 The design answers that question		
2.3 Comparator and outcomes are relevant and defined (measure, time point)		
2.4 The sample size is justified, with its assumptions		

Rating: OK / Concern / Serious concern

3. Pre-specification

Item	Y/N/?	Where
3.1 Registered before the first participant (registry, number, dates)		
3.2 Protocol available		
3.3 SAP available and dated before unblinding or analysis		
3.4 Data and code available, or their absence justified		

Rating: OK / Concern / Serious concern

4. Internal consistency

Item	Y/N/?	Where
4.1 The n’s agree across abstract, flow diagram, tables and text		
4.2 Percentages match their denominators (recompute at least two)		
4.3 CIs and p-values agree (recompute at least one)		
4.4 Summary statistics are possible (GRIM for means of integer data)		
4.5 Abstract, text, tables and figures report the same results		

Rating: OK / Concern / Serious concern

5. External consistency

Item	Y/N/?	Where
5.1 Primary outcome and time point are the same in registry, protocol and paper		
5.2 Analysis follows the SAP, or deviations are reported and justified		
5.3 Blinding, sample size and analysis population match the protocol		
5.4 Other reports of the same study agree; no duplicate publication		
5.5 No PubPeer threads, corrections, expressions of concern or retraction (or they are addressed)		

Rating: OK / Concern / Serious concern

6. Validity

Tool: RoB 2 (randomised), ROBINS-I or ROBINS-E (non-randomised), QUADAS-2 (diagnostic), PROBAST (prediction).

Item	Y/N/?	Where
6.1 Allocation or selection does not favour one group (randomisation, concealment, confounding)		
6.2 Deviations from the intended interventions are unlikely to bias the result		

Item	Y/N/?	Where
6.3 Missing outcome data are few, balanced, or handled appropriately		
6.4 Outcome assessors are blinded, or the outcome is objective		
6.5 The reported result was not selected among several outcomes or analyses		
6.6 Likely direction and size of the main bias are stated (by you)		

Rating: OK / Concern / Serious concern

7. Interpretation

Item	Y/N/?	Where
7.1 The abstract reports the primary outcome and its result		
7.2 No spin: secondary outcomes or subgroups are not promoted over a null primary		
7.3 Causal language only where the design supports it		
7.4 Absolute effects and harms are reported, not only relative effects		
7.5 Generalisation is limited to the population studied		
7.6 Funding and conflicts are declared		

Rating: OK / Concern / Serious concern

Verdict rule

Start at **high** confidence in the main result (**moderate** for a non-randomised design), then apply:

Finding	Effect
Serious concern in layer 1 or 2	The claim is **not supported as stated**: ask for a different claim, not a different analysis
Serious concern in layer 3, 5 or 6	Down **one level** for each
Two or more Concerns across layers 3, 5 and 6	Down **one level**
Serious concern in layer 4	**Very low** until the authors explain; if the pattern suggests fabrication, write to the editor in confidence
Concern in layer 4	Ask for a correction; confidence does not move
Concern or serious concern in layer 7	The **wording** of the conclusion must change (major issue if serious); confidence does not move

- "?" on 3.1, 5.1 or 5.2 counts as a Concern: a missing document is not a pass.

- Levels: high, moderate, low, very low. Copy the result into the review report.
- The recommendation follows from what can be fixed, not from confidence alone: fixable by correction, reanalysis or rewording, major revision; not fixable because the design cannot answer the question, reject.

Layer	1	2	3	4	5	6	7	Confidence
Rating								

Annex: randomised trials (CONSORT 2025)

Use the CONSORT 2025 checklist (consort-spirit.org) for the item numbers. Topics that most often reveal problems, by layer:

Layer	CONSORT topic	What to check
1	Title and abstract	“Randomised” in the title; the abstract gives the primary outcome with its effect and CI
2	Outcomes; sample size	Primary outcome with measure and time point; sample size assumptions match the outcome analysed
3	Trial registration; protocol and SAP; data sharing	Registry number and date of registration versus first enrolment; where the protocol and SAP are
3, 5	Changes to the trial after it started	Any change of outcome, sample size or analysis, with dates and reasons
4	Participant flow; numbers analysed	Flow diagram adds up; each analysis gives its denominator
4	Outcomes and estimation	Each outcome: result per group, effect size, CI; binary outcomes with absolute and relative effects
6	Randomisation: sequence, concealment, implementation	Who generated the sequence, who enrolled, who assigned
6	Blinding	Who was blinded (participants, care providers, outcome assessors) and how
6	Statistical methods	Analysis population (ITT, modified ITT, per protocol), missing data handling, adjustment for stratification factors
6, 7	Harms	Harms collected and reported per group
7	Discussion: limitations, generalisability, interpretation	Interpretation consistent with the primary result; limitations include the biases you found
7	Funding and conflicts	Role of the funder in design, analysis and reporting

AI use (if any)

- ☐ I appraised the paper myself before using AI.
- ☐ I audited every AI claim against the paper or the primary source.
- ☐ No confidential manuscript was uploaded.
- Tool and purpose: _____

Toolkit and Links

Everything you need to apply the 7 layers, grouped by layer.

Layer 1. Article type and reporting guidelines

Study type	Reporting guideline	Where
Randomised trial	CONSORT 2025	consort-spirit.org
Trial protocol	SPIRIT 2025	consort-spirit.org
Crossover trial	CONSORT extension for crossover trials (2019)	EQUATOR
Trial of an AI intervention	CONSORT-AI, SPIRIT-AI (2020)	EQUATOR
Observational study	STROBE	strobe-statement.org
Diagnostic accuracy	STARD 2015	EQUATOR
Prediction model (incl. ML)	TRIPOD+AI (2024)	tripod-statement.org
Systematic review	PRISMA 2020	prisma-statement.org
Any other design	Search the library	equator-network.org

Layer 2. Question and design

- **PICO / PECO:** population, intervention or exposure, comparator, outcome.
- **Estimand** (ICH E9(R1)): treatment, population, variable (endpoint), handling of intercurrent events, and the population-level summary.
- **Target trial emulation:** for observational studies, ask “which randomised trial is this study trying to emulate?”

Layer 3. Pre-specification and open science

Resource	Use
ClinicalTrials.gov	Trial registry. Use “History of changes” to see what was changed and when.
WHO ICTRP	Searches all primary registries at once.
PROSPERO	Registry for systematic reviews.
OSF Registries	Pre-registrations of any study type.
Journal supplements	Protocol and SAP are often in the online supplement (NEJM, Lancet, JAMA, BMJ).

Questions to ask:

- Was registration done **before** the first participant was enrolled?
- Is the SAP dated before unblinding?

Layer 4. Internal consistency: forensic tools

Check	How
Do the n's add up?	Abstract vs flow diagram vs Table 1 vs results tables vs text.
Are percentages right?	Recompute numerator / denominator.
Do the CI and p-value agree?	A 95% CI that excludes the null value (1 for ratios, 0 for differences) means $p < 0.05$, and vice versa. To get p from a ratio CI: SE = (ln(upper) - ln(lower)) / 3.92; z = ln(estimate) / SE; p from z.
Is the mean possible?	**GRIM**: for integer data, mean x n must be (close to) a whole number. Example: n = 15, mean 2.65 gives 39.75; but 39/15 = 2.60 and 40/15 = 2.67, so 2.65 is impossible.
Are the summary statistics plausible?	**SPRITE**: can any dataset produce this mean and SD within the scale limits?
Are the reported test statistics consistent?	statcheck recomputes p from the reported test statistic and degrees of freedom.
Are baseline tables too similar or too different?	Carlisle's method: in a truly randomised trial, baseline p-values are uniformly distributed.

Layer 5. External consistency

Resource	Use
Registry vs paper	Primary outcome, time point, sample size, analysis population.
COMPare	Method for detecting outcome switching.
PubPeer	Post-publication comments, often flagging image or data problems.
Retraction Watch Database	Retractions, corrections and expressions of concern (now via Crossref).
Problematic Paper Screener	Tortured phrases and other paper-mill signals.
PubMed "Comment in" / "Retraction in"	Shown at the top of every PubMed record.

Layer 6. Validity: risk-of-bias tools

Design	Tool	Where
Randomised trial	RoB 2	riskofbias.info
Non-randomised study of an intervention	ROBINS-I	riskofbias.info
Non-randomised study of an exposure	ROBINS-E	riskofbias.info
Diagnostic accuracy	QUADAS-2	bristol.ac.uk/quadas

Design	Tool	Where
Prediction model	PROBAST / PROBAST+AI	probast.org
Body of evidence	GRADE	gradeworkinggroup.org

Biases to look for:

- Confounding.
- Selection, including exclusions after randomisation.
- Information: unblinded outcome assessment, misclassification.
- Immortal time bias.
- Collider bias.
- Missing data.

Layer 7. Interpretation and spin

Spin signals (Boutron et al., JAMA 2010):

- The primary outcome is not significant, but the abstract focuses on secondary outcomes or subgroups.
- A non-significant result is presented as "equivalent" or as a "trend".
- Causal language ("reduces", "prevents") in an observational study.
- The conclusion recommends a practice change that the primary result does not support.

AI rules for reviewers

1. Appraise the paper yourself first, then use the AI, then audit what it produces.
2. Verify every AI claim against the paper or the primary source. LLMs can misread tables and invent registry content and references.
3. **Never upload a confidential manuscript under peer review** to an AI tool. Many journals and funders forbid it; the NIH prohibits it for grant review.
4. Disclose any AI use to the editor, following the journal's policy.

Post-quiz

The same 15 questions are taken at the start (pre) and at the end (post) of the day. One answer per question. About 10 minutes.

Participant code: _____ Post-quiz

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1. A randomised trial finds RR 0.95 (95% CI 0.80 to 1.13) for its primary outcome. The best conclusion is:

- A. The treatment has no effect.
- B. The trial did not show a difference; effects from a 20% reduction to a 13% increase remain compatible with the data.
- C. The treatment is equivalent to control.
- D. The trial was underpowered, so the result should be ignored.

2. A trial's primary outcome in the registry is "mortality at 90 days"; the paper reports "mortality at 28 days" as primary. This is:

- A. Acceptable if the 28-day result is significant.
- B. Outcome switching, which should be disclosed and justified.
- C. A minor reporting issue.
- D. Only a problem if the journal requires registration.

3. The main purpose of a statistical analysis plan (SAP) finalised before unblinding is to:

- A. Calculate the sample size.
- B. Satisfy the ethics committee.
- C. Limit analytical flexibility after the data are seen.
- D. Describe the randomisation procedure.

4. In a trial, 12 randomised patients are excluded from the analysis because they did not receive the intervention. The analysis is described as intention-to-treat. This is:

- A. Correct, because they did not receive the treatment.
- B. Incorrect: intention-to-treat analyses patients in the group they were randomised to.
- C. Correct if the exclusions are balanced between groups.
- D. Correct if the exclusions are reported in the flow diagram.

5. A paper reports OR 1.40 (95% CI 1.05 to 1.87) and p = 0.21. You should:

- A. Trust the p-value.
- B. Trust the CI.
- C. Flag an internal inconsistency: a 95% CI excluding 1 implies $p < 0.05$.
- D. Ignore it; both are correct with different tests.

6. In a subgroup of n = 17 patients, the mean number of previous pregnancies (an integer variable) is reported as 2.30. This is:

- A. Plausible.
- B. Impossible, because 2.30 x 17 = 39.1 is not close to a whole number.
- C. Only checkable with the raw data.
- D. Possible if some values are missing.

7. Which design can best support a causal claim about a drug's effect?

- A. A large registry cohort with a very small p-value.
- B. A well-conducted randomised controlled trial.
- C. A case series of consecutive patients.
- D. A cross-sectional survey with adjustment for 50 covariates.

8. The reporting guideline for a study developing a machine-learning prediction model is:

- A. CONSORT-AI
- B. STARD
- C. TRIPOD+AI
- D. STROBE

9. The risk-of-bias tool for a non-randomised study comparing two treatments is:

- A. RoB 2
- B. ROBINS-I
- C. QUADAS-2
- D. GRADE

10. The primary outcome is non-significant, and the abstract conclusion highlights a significant subgroup. This is an example of:

- A. Good transparency.
- B. Spin.
- C. Multiplicity correction.
- D. A pre-specified analysis, so not a problem.

11. Which is a typical signal of a paper-mill product?

- A. Data shared on a public repository.
- B. Tortured phrases such as “counterfeit consciousness” for “artificial intelligence”.
- C. A long limitations section.
- D. Registration before enrolment.

12. You are reviewing a confidential manuscript for a journal. Regarding AI tools:

- A. You may upload the full manuscript to any LLM to save time.
- B. You must not upload the confidential manuscript; follow the journal’s policy and disclose any permitted use.
- C. You may upload it if you delete the chat afterwards.
- D. AI use is never relevant to peer review.

13. An LLM tells you that a trial’s registry entry lists a different primary outcome. The best next step is:

- A. Report it as a major issue.
- B. Ignore it.
- C. Open the registry yourself and verify, including the history of changes.
- D. Ask the LLM again to confirm.

14. Unblinded outcome assessment is most likely to bias:

- A. All-cause mortality.
- B. A subjective outcome such as a functional score rated by the treating team.
- C. Neither: randomisation protects against it.
- D. Only the baseline characteristics.

15. In a review report, a major issue should ideally contain:

- A. Your general impression of the authors’ competence.
- B. The problem, why it matters, and a concrete request the authors can act on.
- C. Only the recommendation to reject.
- D. A list of typos.